Ehtimollik modellari: N-grammalar va so'z turkumlarini teglash.

Tabiiy tilni qayta ishlash — 5-ma'ruza

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11:00–12:20 · mavzu №5

23-iyun 2026

Prezentatsiya rejasi

Takrorlash: o'tgan darslardan ikki savol

1-savol

4-ma'ruzadagi shovqinli kanal (*noisy channel*) modelida

$$\hat{w} = \arg\max_{w} P(x \mid w)\, P(w)$$

edi. Bu yerdagi til modeli $P(w)$ — bitta so'z ehtimoli — **nechanchi tartibli** N-gram?

2-savol

4-amaliyotda `edit_distance` dinamik dasturlash (*Dynamic Programming, DP*) bilan hisoblangan edi. Bugungi Viterbi ham DP. Sizningcha, nega ikkalasi ham DP?

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1-javob

Unigram (1-gram) — so'z chastotasi, kontekstsiz. Bugun uni bigram va trigramga kengaytiramiz, ya'ni kontekstni hisobga olamiz.

2-javob

Ikkalasi ham masalani kichik qism-masalalarga bo'lib, ularning optimal yechimidan butun yechimni quradi — bu DP mohiyati.

Bugungi dars maqsadlari

1. Bigram ehtimolini chastotani sanash asosida — maksimal ehtimollik bahosi (*Maximum Likelihood Estimation, MLE*) — zanjir qoidasi (*chain rule*) va Markov taxminidan (*Markov assumption*) **keltirib chiqara olasiz**.
2. Add-1 (Laplace) silliqlash (*smoothing*) va perplexity (modelning matndan *taajjublanish* darajasi) ni kichik korpusda qo'lda **hisoblay olasiz**.
3. Markov zanjiri bilan so'z turkumi (*Part-of-Speech, POS*) teglari ketma-ketligi ehtimolini **qo'llay olasiz**.
4. Viterbi algoritmini dinamik dasturlash (DP) bilan ishlatib, eng ehtimoliy teg ketma-ketligini **keltirib chiqara olasiz** — jumladan Viterbi jadvalidagi bitta qiymatni qo'lda topasiz.

Oldingi bilimlar: 2-ma'ruzadan ehtimollik, $\arg\max$ va logarifm; 4-ma'ruzadan dinamik dasturlash va til modeli $P(w)$; matematikadan ko'paytma va maksimum.

Bugungi dars rejasi

Tsikl	Mavzu	Natija
1	N-gram til modeli: unigram, bigram, trigram	$P(\text{kitob} \mid \text{men}) = 2/3$
2	Laplace silliqlash, keyin perplexity	add-1 va PP
3	POS teglash va Markov zanjiri	$P(\mathrm{NN}, \mathrm{VB}) = 0.42$
4	HMM va Viterbi algoritmi	$\delta(\mathrm{VB}, t{=}2) = 0.3402$
–	Xulosa va 5-amaliyotga ko'prik	5-amaliyot yo'nalishi

Bu dars 2-haftani ochadi. Klassik quvurdan (1-hafta) ehtimollik modellari (2-hafta) ga o'tamiz. Bugungi Viterbi qiymati $\delta(\mathrm{VB}, t{=}2) = 0.3402$ ertangi 5-amaliyotda `assert` sifatida tekshiriladi. Natija ustunidagi sonlar — soddalashtirilgan o'quv misolidan.

Bugungi darsda yechiladigan muammo: `men` . . . dan keyin qaysi so'z?

Ikki vazifa

1. Avtomatik to'ldirish

Telefon klaviaturasi "`men`" dan keyin keyingi so'zni taklif qiladi. Qaysi so'z eng ehtimoliy?

2. So'z turkumini aniqlash

"`yozdi`" — fe'lmi yoki ot? Buni kontekst hal qiladi. Butun gapdagi **teglar ketma-ketligini** qanday topamiz?

Bugungi yechimlar

1. **N-gram til modeli**: oldingi so'z(lar)ga qarab keyingisini ehtimollik bilan bashorat qiladi.
2. **Yashirin Markov modeli** (*Hidden Markov Model, HMM*) + **Viterbi**: yashirin teglar ketma-ketligining eng ehtimoliysini topadi.

Ikkala yechimning yuragida bitta g'oya: **ehtimollik** va uni chastotani sanash asosida baholash.

Prezentatsiya rejasi

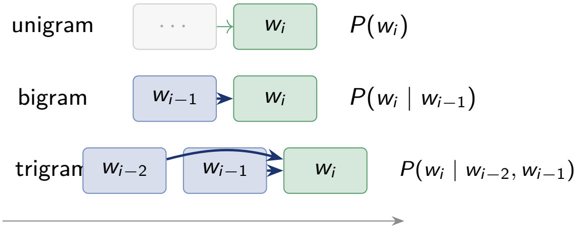
Intuitsiya: keyingi so'zni oldingilariga qarab bashorat qilamiz

`"men"` $\rightarrow$ `kitob? non? keldim?`

Til modeli (*language model*) har bir davom uchun **ehtimol** beradi va eng yuqorisini taklif qiladi.

Nechta oldingi so'z hisobga olinadi?

- **unigram**: kontekstsiz — $P(w)$
- **bigram**: 1 oldingi — $P(w_i \mid w_{i-1})$
- **trigram**: 2 oldingi — $P(w_i \mid w_{i-2}, w_{i-1})$



Ko'proq kontekst $\Rightarrow$ aniqroq bashorat, lekin ko'proq ma'lumot kerak (kam uchraydigan N-gramlar). Bu murosani 2-tsiklda ko'ramiz.

N-gram til modeli: rasmiy ta'rif

$$P(w_i \mid w_{i-1}) = \frac{\text{count}(w_{i-1}, w_i)}{\text{count}(w_{i-1})}$$

$$\text{count}(w_{i-1}, w_i)$$

w_{i-1} dan keyin w_i kelgan hollar soni.

MLE (*Maximum Likelihood Estimation*) — chastotaga asoslangan baho.

Gap ehtimoli (bigram):

$$P(w_1, \ldots, w_n) \approx \prod_{i=1}^{n+1} P(w_i \mid w_{i-1})$$

$$w_0 = \langle s \rangle, \qquad w_{n+1} = \langle / s \rangle$$

$\langle s \rangle$ — gap boshi, $\langle / s \rangle$ — gap oxiri.

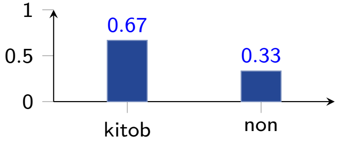
Misol: `men` **dan keyingi so'zlar**

`kitob: 2` `non: 1`

$\text{count}(\texttt{men}) = 3$

shu songa bo'lamiz

$P(\cdot \mid \texttt{men})$



$$P(\texttt{kitob} \mid \texttt{men}) = \frac{2}{3} \approx 0.67$$

$$P(\texttt{non} \mid \texttt{men}) = \frac{1}{3} \approx 0.33$$

Amaliy eslatma: gap chegaralari va maxsus tokenlar

N-gram modelda gap boshi va oxiri ham hisobga olinadi. Shunda birinchi so'z ham ehtimollik oladi:

$$P(\text{men} \mid \langle s \rangle).$$

Gap oxiri $\langle / s \rangle$ ham zarur: usiz model qisqa gaplarga moyil bo'ladi va ehtimollar taqsimoti noto'liq qoladi.

$\langle s \rangle$ → `men` → `kitob` → `o'qidim` → $\langle / s \rangle$

O'zbek matni uchun. Apostrof, tire, raqam va qo'shimchalar normallashtirilmasa, sanoqlar sun'iy bo'linib ketadi: `o'qidim` va `o qidim` ikki xil token bo'lib qolishi mumkin.

Amalda gap boshi `<s>`, gap oxiri `</s>` va noma'lum so'z `<UNK>` (*unknown*) belgilari izchil qo'llanadi.

Keltirib chiqarish: zanjir qoidasidan bigramga

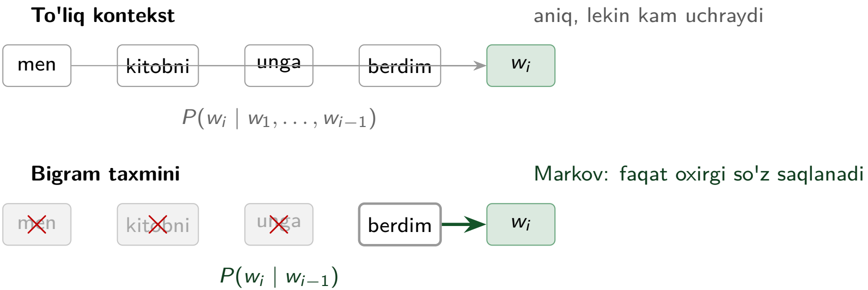
1-qadam. Zanjir qoidasi (*chain rule*) — ehtimolliklar ko'paytmasi:

$$P(w_1, \ldots, w_n) = \prod_{i=1}^{n} P(w_i \mid w_1, \ldots, w_{i-1})$$

2-qadam. Muammo: bunday uzun kontekstlar kam uchraydi, shuning uchun ularni ishonchli sanash qiyin.

3-qadam. **Markov taxmini** (*Markov assumption*): faqat oxirgi $n-1$ ta so'z muhim deb olinadi. Bigramda bu:

$$P(w_i \mid w_1, \ldots, w_{i-1}) \approx P(w_i \mid w_{i-1})$$



Qo'lda misol: bigram ehtimolini sanaymiz

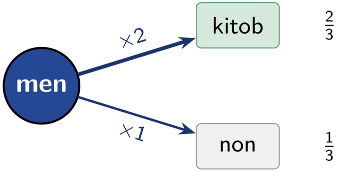
Korpus: S1: men kitob o'qidim S2: men non yedim S3: men kitob oldim

Sanaymiz

$$\text{count}(\text{men}) = 3, \qquad \text{count}(\text{men}, \text{kitob}) = 2$$

$$P(\text{kitob} \mid \text{men}) = \tfrac{2}{3} \approx 0.667$$

Lug'at hajmi $|V| = 6$: men, kitob, o'qidim, non, yedim, oldim.



"men" dan keyin `kitob` (2 marta) `non` (1 marta) dan ko'proq kutiladi ⇒ avtomatik to'ldirish `kitob` ni taklif qiladi.

Sizning vazifangiz: yana bir bigram

Vazifa

Xuddi shu korpus (S1, S2, S3) uchun:

$$P(\text{non} \mid \text{men}) = ?$$

Sizning vazifangiz: yana bir bigram

Vazifa

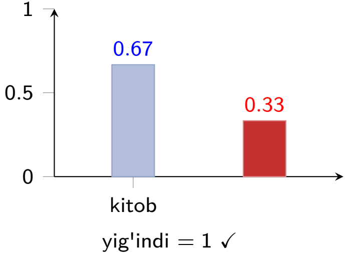
$$P(\text{non} \mid \text{men}) = ?$$

Javob

$\text{count}(\text{men}, \text{non}) = 1, \ \text{count}(\text{men}) = 3$

$$P(\text{non} \mid \text{men}) = \tfrac{1}{3} \approx 0.333$$

Tekshiruv: $\frac{2}{3} + \frac{1}{3} = 1$ — "men" dan keyin faqat shu ikki so'z kelgan.



Kod va formula: bigram modeli

Python kodi

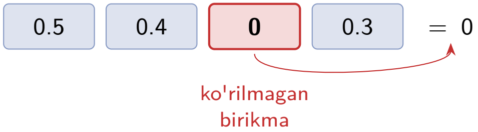
```
from collections import Counter
corpus = [["men","kitob","o'qidim"],
          ["men","non","yedim"],
          ["men","kitob","oldim"]]
uni = Counter(w for s in corpus for w in s)
bi  = Counter((s[i], s[i+1])
              for s in corpus
              for i in range(len(s)-1))

def p_bigram(prev, w):
    return bi[(prev, w)] / uni[prev]

p = p_bigram("men", "kitob")
print(round(p, 3))   # 0.667
```

Kod va formula mosligi

Kod	Formula
`bi[(prev,w)]`	$\text{count}(w_{i-1}, w_i)$
`uni[prev]`	$\text{count}(w_{i-1})$
`bi/uni`	$P(w_i \mid w_{i-1})$

Ertangi 5-amaliyotdagi `m05 Autocomplete` moduli aynan shu sanashga tayanadi.

N-gram tuzog'i: ko'rilmagan birikma butun gapni nolga aylantiradi



Nol muammosi (*zero problem*)

Test gapi: `"men non o’qidim"`.

`(non, o’qidim)` korpusda hech uchramagan $\Rightarrow P = 0$.

Bigram ko'paytmada bitta 0 $\Rightarrow$ **butun gap ehtimoli** $= 0$.

Model ko'rmagan birikmani “imkonsiz” deb hisoblaydi — til esa cheksiz ijodiy. **2-tsikl ikki qadam:** (1) **silliqlash** nol muammosini hal qiladi; (2) keyin tayyor modelni **perplexity** bilan baholaymiz.

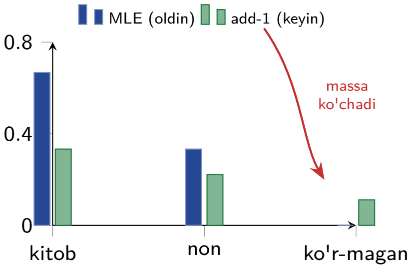
Prezentatsiya rejasi

Add-1 (Laplace) silliqlash: ta'rif

Nol muammosi yodimizda: ko'rilmagan birikma $\Rightarrow P = 0$. Eng oddiy yechim — har bigram sanog'iga $+1$ (*add-1* / *Laplace*):

$$P_{+1}(w_i \mid w_{i-1}) = \frac{\text{count}(w_{i-1}, w_i) + 1}{\text{count}(w_{i-1}) + |V|}$$

Ko'rilgan birikmalardan ehtimollik massasining bir qismi ko'rilmaganlarga **ko'chadi** — model "yangilik"ka tayyor bo'ladi. Maxrajdagi $|V|$ ning sababini keyingi slaydda chiqaramiz.



ko'rilgan ↓, ko'rilmagan 0→>0

Keltirib chiqarish: nega maxrajga $|V|$ qo'shamiz?

1-qadam. Har bigram sanog'iga $+1$ qo'shamiz; shunda ko'rilmaganlar ham 0 emas, 1 bo'ladi.
2-qadam. Ehtimollar yig'indisi 1 bo'lishi shart. w_{i-1} dan keyin har bir lug'at so'zi ($|V|$ ta) uchun $+1$ qo'shdik:

$$\sum_{w} \big(\text{count}(w_{i-1}, w) + 1\big) = \text{count}(w_{i-1}) + |V|$$

3-qadam. Shuning uchun maxraj $\text{count}(w_{i-1}) + |V|$.

$+1$	$+1$	$+1$	$+1$	$+1$	$+1$
kitob	non	men	yedim	oldim	o'qidim

$|V| = 6$ ta so'z, har biriga $+1$ $\Rightarrow$ maxrajga $+|V|$

Qo'lda misol: silliqlash nol muammosini hal qiladi

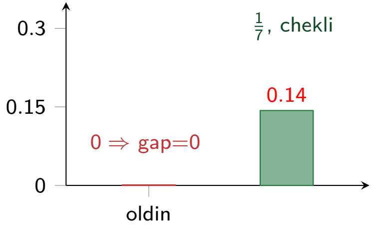
$|V| = 6,\ \text{count}(\text{non}) = 1,\ \text{count}(\text{non}, \text{o'qidim}) = 0.$

Silliqlashsiz:

$$P(\text{o'qidim} \mid \text{non}) = \tfrac{0}{1} = 0 \ \Rightarrow\ \text{gap ehtimoli} = 0$$

Add-1 bilan:

$$P_{+1}(\text{o'qidim} \mid \text{non}) = \tfrac{0+1}{1+6} = \tfrac{1}{7} \approx 0.143$$



Silliqlash modelni "yangilik"ka tayyor qiladi.

Sizning vazifangiz: add-1 ehtimolini hisoblang

Vazifa

$|V| = 6,\ \text{count}(\text{men}) = 3,\ \text{count}(\text{men}, \text{kitob}) = 2$:

$$P_{+1}(\text{kitob} \mid \text{men}) = ?$$

Silliqlashsiz qiymat 2/3 edi — add-1 dan keyin qanday oʻzgaradi?

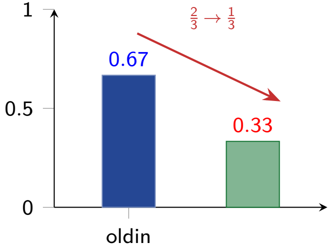
Vazifa

$$P_{+1}(\text{kitob} \mid \text{men}) = ?$$

Javob

$$P_{+1}(\text{kitob} \mid \text{men}) = \frac{2+1}{3+6} = \frac{3}{9} = \tfrac{1}{3} \approx 0.333$$

Ko'rilgan birikma ehtimoli $2/3 \to 1/3$ ga pasaydi — massa ko'rilmaganlarga ko'chdi. Bu silliqlash narxi.



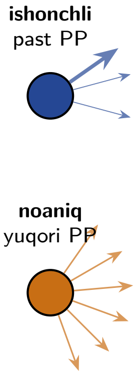
Perplexity (PP) nima?

Perplexity — model test matnini ko'rganda qanchalik noaniq qaror qilayotganini ko'rsatadigan baholash mezoni.

- **Past PP** — model keyingi tokenlarni yaxshi kutgan.
- **Yuqori PP** — model ko'p variant orasida noaniq qolgan.

$$\mathrm{PP}(W) = \exp\left(-\frac{1}{N}\sum_{i=1}^{N} \ln P(w_i \mid w_{i-1})\right)$$

N — test matndagi tokenlar soni. Bigram modelda har bir token ehtimoli oldingi token orqali baholanadi.



PP taxminan model har qadamda nechta teng ehtimolli variant orasidan tanlayotgandek ekanini bildiradi.

Kod va formula: add-1 va perplexity

Python kodi

```
import math
V = len(uni)                  # lug'at hajmi = 6

def p_add1(prev, w):
    return (bi[(prev, w)] + 1) / (uni[prev] + V)

def perplexity(pairs):      # pairs: [(prev,w),...]
    log_sum = sum(math.log(p_add1(a, b))
                  for a, b in pairs)
    return math.exp(-log_sum / len(pairs))

pa = p_add1("non", "o'qidim")
print(round(pa, 3))   # 0.143
```

Kod va formula mosligi

Kod	Formula
`+ 1`	$+1$ (Laplace)
`+ V`	$+\lvert V\rvert$
`-log_sum/N`	$-\frac{1}{N}\sum \ln P$
`exp(...)`	PP

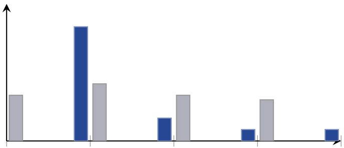
Log fazoda yig'amiz — to'g'ridan-to'g'ri ko'paytma arifmetik kichrayishga (*underflow*) olib keladi (4-tsikl).

Perplexity tuzog'i: nimani taqqoslayapsiz?

Cheklovlar

- Perplexity faqat **bir xil lug'at**, **bir xil tokenizatsiya** va **bir xil test to'plam**da taqqoslanadi.
- Lug'atdan tashqari (*Out-Of-Vocabulary, OOV*) so'zlar perplexity ni keskin buzadi.
- Ortiqcha silliqlash taqsimotni tekislaydi — model "hammasi mumkin" deydi, foydasi kamayadi.

kam silliqlash vs ortiqcha silliqlash



ko'k: ishonchli kulrang: tekis (foydasiz)

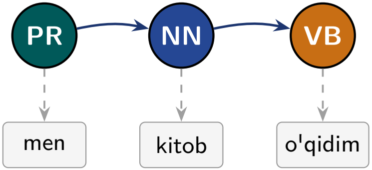
Past perplexity — **vosita**, maqsad emas. Yakuniy vazifa sifati (to'ldirish to'g'riligi) muhimroq. Yaxshiroq usullar: add-k (kichikroq $+k$), *backoff* (kam uchrasa qisqaroq N-gramga tushadi), Kneser–Ney.

Prezentatsiya rejasi

So'z turkumini aniqlash (*Part-of-Speech, POS*): har so'zga **teg** — ot (NN), fe'l (VB), sifat...

Markov g'oyasi: keyingi teg asosan *oldingi* tegga bog'liq. Otdan keyin ko'pincha fe'l, sifatdan keyin ot...

Bir xil so'z turli teg olishi mumkin: **kontekst** hal qiladi. Markov zanjiri teglar orasidagi **o'tishlar**ni modellaydi.



teglar (yuqorida) **zanjir** hosil qiladi; so'zlar (pastda) — kuzatiladigan natija

PR — olmosh, NN — ot, VB — fe'l

Markov zanjiri: rasmiy ta'rif

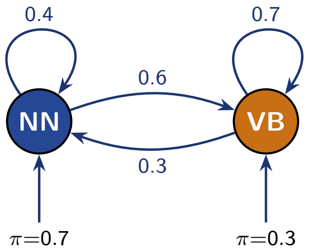
$$P(s_1 s_2 \dots s_T) = \pi(s_1) \prod_{t=2}^{T} A(s_t \mid s_{t-1})$$

s_t — t-holat (teg); $\pi(s_1)$ — boshlang'ich ehtimol; $A(s_t \mid s_{t-1})$ — o'tish (*transition*) ehtimoli. 1-tartibli zanjir: keyingi holat faqat *bitta* oldingisiga bog'liq. Har bir satr yig'indisi 1: $\sum_j A(j \mid i) = 1$ (shartli taqsimot).

O'tish matritsasi A (soddalashtirilgan o'quv misoli):

	$\to$ NN	$\to$ VB
NN	0.4	0.6
VB	0.3	0.7

$\pi(\text{NN}) = 0.7, \quad \pi(\text{VB}) = 0.3$



o'qlar — o'tish ehtimollari; pastdan — π (boshlanish)

1-qadam. Teglar ketma-ketligining to'liq ehtimoli (zanjir qoidasi):

$$P(s_1 \dots s_T) = \pi(s_1) \prod_{t=2}^{T} P(s_t \mid s_1 \dots s_{t-1})$$

2-qadam. 1-tartibli Markov taxmini — faqat oldingi holat muhim:

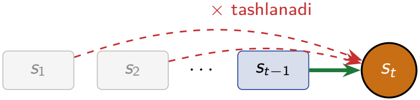
$$P(s_t \mid s_1 \dots s_{t-1}) \approx P(s_t \mid s_{t-1}) = A(s_t \mid s_{t-1})$$

3-qadam. Demak:

$$P(s_1 \dots s_T) = \pi(s_1) \prod_{t=2}^{T} A(s_t \mid s_{t-1})$$

Bu — N-gram bilan bir xil g'oya, faqat so'zlar o'rniga **teglar** ustida.

faqat oxirgi teg muhim



yashil — saqlanadi;
qizil — e'tiborsiz qoldiriladi

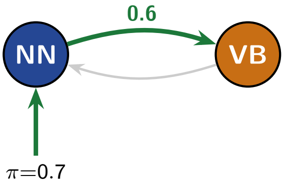
Qo'lda misol: teg ketma-ketligi ehtimoli

$\pi(\text{NN}) = 0.7, \quad A(\text{VB} \mid \text{NN}) = 0.6$ (soddalashtirilgan o'quv misoli).

Hisoblaymiz:

$$P(\text{NN}, \text{VB}) = \pi(\text{NN}) \cdot A(\text{VB} \mid \text{NN})$$
$$= 0.7 \times 0.6 = \mathbf{0.42}$$

Bu faqat *teglar* ehtimoli (so'zlarsiz). Keyingi tsiklda so'zlar (kuzatuvlar) qo'shilib, HMM hosil bo'ladi.



$\boxed{0.7 \times 0.6 = 0.42}$

boshlash → NN, so'ng NN → VB

Sizning vazifangiz: teskari ketma-ketlik

Vazifa

$\pi(\mathrm{VB}) = 0.3, \quad A(\mathrm{NN} \mid \mathrm{VB}) = 0.3.$

$$P(\mathrm{VB}, \mathrm{NN}) = ?$$

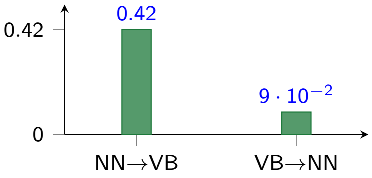
Vazifa

$\pi(\mathrm{VB}) = 0.3$, $A(\mathrm{NN} \mid \mathrm{VB}) = 0.3$; $P(\mathrm{VB}, \mathrm{NN}) = ?$

Javob

$$P(\mathrm{VB}, \mathrm{NN}) = \pi(\mathrm{VB}) \cdot A(\mathrm{NN} \mid \mathrm{VB})$$
$$= 0.3 \times 0.3 = 0.09$$

$0.09 < 0.42$ — bu korpusda gap ot bilan boshlanib fe'l bilan davom etishi ancha ehtimoliyroq (NN→VB).



0.42 — ehtimoliyroq teg tartibi

Kod va formula: Markov ketma-ketligi

Python kodi

```
pi = {"NN": 0.7, "VB": 0.3}
A = {("NN","NN"): 0.4, ("NN","VB"): 0.6,
     ("VB","NN"): 0.3, ("VB","VB"): 0.7}

def seq_prob(tags):
    p = pi[tags[0]]
    for i in range(1, len(tags)):
        p *= A[(tags[i-1], tags[i])]
    return p

print(round(seq_prob(["NN","VB"]), 2))  # 0.42
print(round(seq_prob(["VB","NN"]), 2))  # 0.09
```

Kod va formula mosligi

Kod	Formula
`pi[tags[0]]`	$\pi(s_1)$
`A[(prev,cur)]`	$A(s_t \mid s_{t-1})$
`p *= ...`	$\prod$

A — o'tish matritsasi (lug'at sifatida). 5-amaliyotdagi `m05b` `POSTagger` moduli shuni kengaytiradi.

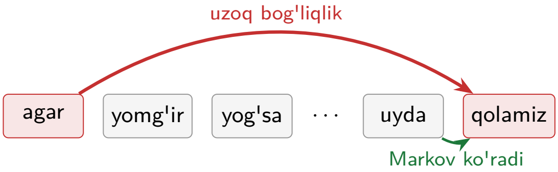
Markov tuzog'i: uzoq bog'liqlik ko'rinmaydi

1-tartib cheklovi

Keyingi holat faqat *bitta* oldingisiga bog'liq. Lekin:

"agar yomg'ir yog'sa... biz uyda qolamiz"

*"qolamiz" so'zi "agar"ga bog'liq — ammo ular orasida ko'p so'z bor. 1-tartibli Markov buni **ko'rmaydi**.*



Yuqori tartib (2-, 3-) biroz yordam beradi.
To'liq yechim: **LSTM** va **Transformer** (keyingi haftalarda).

O'zbek POS teglashda amaliy muammolar

O'zbek tilida bitta so'z shakli ko'p grammatik ma'lumot tashiydi:

- `kitoblarimizdan`: ot + ko'plik + egalik + kelishik;
- `yozgan`: fe'ldan yasalgan sifatdosh shakl; gapda sifatga yaqin vazifada kelishi mumkin;
- erkinroq so'z tartibi teglar ketma-ketligini oldindan taxmin qilishni murakkablashtiradi.

kitob	lar	imiz	dan
ot	ko'plik	egalik	kelishik

bitta so'z — to'rt morfema

Amaliy yechimlar

HMM uchun **teglangan korpus** zarur. Korpus kichik bo'lsa:

- morfologik analizator orqali mumkin bo'lgan teglar toraytiriladi;
- qoida va lug'atlar bilan xatolar kamaytiriladi;
- qo'lda tekshirilgan namunalar bilan model baholanadi;
- keyinroq neyron modellar bilan gibrid yondashuv quriladi.

Prezentatsiya rejasi

Markov zanjiri (3-bo'lim): teglar **ko'rinadi**. **HMM**: teglar **yashirin** — faqat so'zlar ko'rinadi.

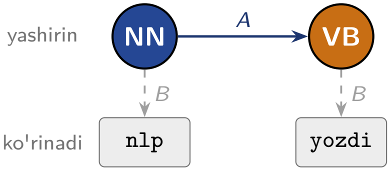
Biz **so'zlar**ni ko'ramiz (`nlp`, `yozdi`), lekin **teglar** (NN, VB) yashirin. Yashirin Markov modeli (HMM):

- teglar Markov zanjiri bo'ylab o'tadi (A — o'tish);
- har teg so'zni "chiqaradi" (B — chiqarish, *emission*).

π, A, B — chastotadan (MLE) o'rganiladi (teglangan korpusdan).

Vazifa: so'zlar berilgan, eng ehtimoliy teg ketma-ketligini topish.

Barcha teg yo'llarini sinash eksponensial (S^T). **Viterbi** — DP bilan eng yaxshi yo'lni $O(T \cdot S^2)$ da topadi.



teglar zanjiri (A) so'zlarni chiqaradi (B)

HMM (*Hidden Markov Model* — yashirin Markov modeli) ko'rinadigan so'zlar ortida yashirin POS teglar ketma-ketligi bor deb qaraydi.

Model qismlari

- $\pi(j)$ — gap boshida j teg kelish ehtimoli;
- $A(j \mid i)$ — i tegdan j tegga o'tish ehtimoli;
- $B(o_t \mid j)$ — j tegda o_t so'zini ko'rish ehtimoli.

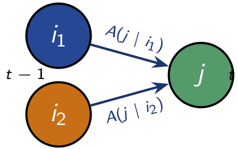
Viterbi — barcha teg ketma-ketliklarini sinamay, dinamik dasturlash orqali eng ehtimoliy yo'lni topadi.

Asosiy rekurrensiya

$$\delta_t(j) = \left[\max_i \delta_{t-1}(i) A(j \mid i)\right] B(o_t \mid j)$$

Formula talqini

- $\delta_t(j)$ — t-pozitsiyada j teg bilan tugaydigan eng yaxshi yo'l balli;
- $\max_i$ — oldingi teglardan eng yaxshisini tanlash;
- $B(o_t \mid j)$ — hozirgi so'zning j tegga mosligi.



eng yaxshi kirish tanlanadi
va $B(o_t \mid j)$ bilan ko'paytiriladi

Keltirib chiqarish: nega Viterbi DP ishlaydi?

1-qadam. Maqsad — $\arg\max_s P(s \mid o)$; ammo o (so'zlar) berilganda $P(o)$ s ga bog'liq emas, shuning uchun qo'shma ehtimolni maksimallashtiramiz: $\arg\max_{s_1\ldots s_T} P(s_1\ldots s_T, o_1\ldots o_T)$. Barcha yo'llar — S^T ta (juda ko'p).

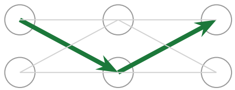
2-qadam. Kuzatuv: t da j holatda tugaydigan eng yaxshi yo'l faqat $t{-}1$ dagi eng yaxshi yo'llarga tayanadi (Markov xossasi).

3-qadam. Shuning uchun har qadamda har holat uchun eng yaxshi kirishni saqlaymiz:

$$\delta_t(j) = \left[\max_i \delta_{t-1}(i) A(j \mid i)\right] B(o_t \mid j)$$

Eksponensial qidiruv $O(TS^2)$ ga tushadi — 4-ma'ruzadagi `edit_distance` bilan bir xil DP mantiqi.

S^T yo'l ⇒ qadamli max



Viterbi **panjarasi** (*trellis*); yashil — saqlangan eng yaxshi yo'l

Qo'lda hisob: Viterbi — $\delta(\mathrm{VB}, t{=}2) = 0.3402$

Parametrlar (soddalashtirilgan o'quv misoli): $\pi(\mathrm{NN}){=}0.7$, $\pi(\mathrm{VB}){=}0.3$;
$B(\text{nlp}|\mathrm{NN}){=}0.9$, $B(\text{nlp}|\mathrm{VB}){=}0.1$, $B(\text{yozdi}|\mathrm{NN}){=}0.1$, $B(\text{yozdi}|\mathrm{VB}){=}0.9$;
$A(\mathrm{VB}|\mathrm{NN}){=}0.6$, $A(\mathrm{NN}|\mathrm{VB}){=}0.3$, $A(\mathrm{NN}|\mathrm{NN}){=}0.4$, $A(\mathrm{VB}|\mathrm{VB}){=}0.7$

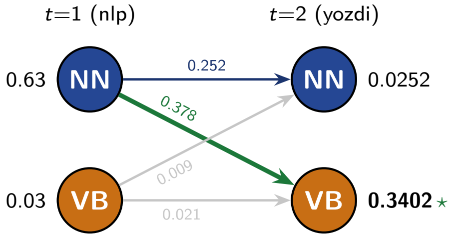
$t{=}1$ **(nlp):**

$$\delta_1(\mathrm{NN}) = 0.7 \cdot 0.9 = 0.63$$
$$\delta_1(\mathrm{VB}) = 0.3 \cdot 0.1 = 0.03$$

$t{=}2$ **(yozdi), VB:**

$$\delta_2(\mathrm{VB}) = \max(0.63 \cdot 0.6,\ 0.03 \cdot 0.7) \cdot 0.9 = 0.378 \cdot 0.9 = \mathbf{0.3402}$$

Ertangi 5-amaliyotning birinchi assert'i:
`abs(delta_VB_t2 - 0.3402) < 1e-4`



yashil — eng ehtimoliy yo'l; orqaga kuzatish: VB ← NN ⇒ **[NN, VB]**

Sizning vazifangiz: $\delta_2(\mathrm{NN})$ ni hisoblang

Vazifa

Xuddi shu HMM uchun $t = 2$ (*yozdi*) da NN qiymatini hisoblang.

$$\delta_2(\mathrm{NN}) = \max \begin{cases} \delta_1(\mathrm{NN}) A(\mathrm{NN} \mid \mathrm{NN}), \\ \delta_1(\mathrm{VB}) A(\mathrm{NN} \mid \mathrm{VB}) \end{cases} \cdot B(\text{yozdi} \mid \mathrm{NN})$$

$$\delta_1(\mathrm{NN}) = 0.63, \quad \delta_1(\mathrm{VB}) = 0.03$$

$$A(\mathrm{NN} \mid \mathrm{NN}) = 0.4, \quad A(\mathrm{NN} \mid \mathrm{VB}) = 0.3, \quad B(\text{yozdi} \mid \mathrm{NN}) = 0.1$$

Vazifa

$$\delta_2(\mathrm{NN}) = ?$$

1-qadam. NN ga keladigan ikki yo'lni hisoblaymiz

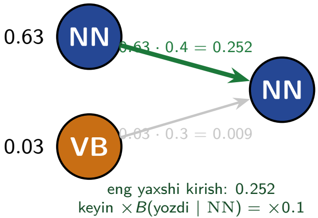
$$\mathrm{NN} \to \mathrm{NN}: \quad 0.63 \cdot 0.4 = 0.252$$

$$\mathrm{VB} \to \mathrm{NN}: \quad 0.03 \cdot 0.3 = 0.009$$

2-qadam. Eng katta kirishni tanlaymiz

$$\max(0.252,\ 0.009) = 0.252$$

3-qadam. *yozdi* ning NN bo'yicha ehtimoliga ko'paytiramiz



$$\boxed{\delta_2(\mathrm{NN}) = 0.0252}$$

Bu faqat $t = 2$ dagi **NN** katagi uchun hisob.

Oldingi hisobda $\delta_2(\mathrm{VB}) = 0.3402$ bo'lsa, $0.3402 > 0.0252$, demak $t = 2$ da VB kuchliroq.

Kod va formula: Viterbi

Python kodi

```
def viterbi(obs, states, pi, A, B):
    delta = [{s: pi[s]*B[(obs[0],s)] for s in states}]
    back = [{}]
    for t in range(1, len(obs)):
        d, bk = {}, {}
        for j in states:
            i = max(states,
                    key=lambda i: delta[t-1][i]*A[(i,j)])
            d[j] = delta[t-1][i]*A[(i,j)]*B[(obs[t],j)]
            bk[j] = i
        delta.append(d); back.append(bk)
    last = max(states, key=lambda s: delta[-1][s])
    path = [last]
    for t in range(len(obs)-1, 0, -1):
        last = back[t][last]; path.insert(0, last)
    return path, delta
```

Kod va formula mosligi

Kod	Formula
`pi[s]*B`	$\delta_1(j)$
`max(...A)`	$\max_i \delta A$
`*B[(o,j)]`	$\cdot B(o_t \mid j)$
`bk[j]=i`	$\psi_t(j)$

`path == ["NN","VB"]`; `delta[1]["VB"]` $\approx$ 0.3402. 5-amaliyotdagi `m05b POSTagger` shuni amalga oshiradi.

Viterbi tuzog'i: ehtimollar ko'paytmasi kichrayadi

Muammo: underflow

Uzun ketma-ketlikda kichik ehtimollar ketma-ket ko'paytiriladi:

$$0.6^{50} \approx 8.1 \cdot 10^{-12}$$

Son juda kichrayganda kompyuter uni **0** deb saqlashi mumkin.

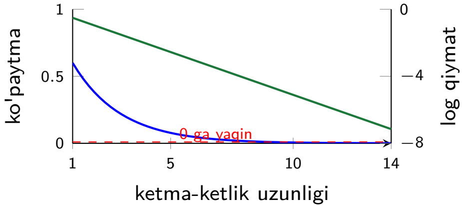
Yechim: log fazo

Ko'paytma logarifmlar yig'indisiga aylantiriladi:

$$\log(ab) = \log a + \log b$$

Viterbi amalda ko'pincha **log-ehtimollar** bilan hisoblanadi.

Ko'paytma va log-ko'paytma farqi



Ko'paytma tez 0 ga yaqinlashadi; log qiymat esa tekis kamayadi va hisoblash uchun qulay.

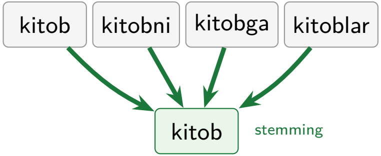
Prezentatsiya rejasi

O'zbek tilida: SOV tartibi va agglutinatsiya N-gramni qiynaydi

1. SOV tartibi (ega–to'ldiruvchi–kesim, *Subject–Object–Verb*; kesim oxirida):

Kesim (`berdim`) oxirida. Bigram "men" keyingisini taxmin qiladi, lekin ma'noni belgilovchi fe'l **uzoqda** $\Rightarrow$ uzoq bog'liqlik.

2. Agglutinatsiya (*agglutination* — qo'shimchalar ketma-ket ulanishi):



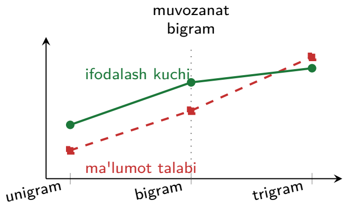
Oqibat: $|V|$ portlaydi; ko'p N-gramlar 0/1 marta uchraydi $\Rightarrow$ **siyrak** ma'lumot, silliqlash ayniqsa muhim. **Yechim:** stemming (*stemming* — so'zni o'zagiga keltirish) N-gramdan oldin $|V|$ ni kichraytiradi.

Sintez: kontekst oshsa, ma'lumot talabi ham oshadi

Model	Kontekst	Murosa
Unigram	yo'q	sodda, lekin kontekst yo'q
Bigram	1 oldingi	amaliy muvozanat
Trigram	2 oldingi	aniqroq, ammo siyrakroq
HMM	yashirin teglar	ketma-ketlik teglash

Asosiy g'oya

Asosiy mexanizm: **ehtimollikni sanash** + **Markov taxmini** + kerakli joyda **silliqlash**. Viterbi shu g'oyani POS teglar ketma-ketligini topishga qo'llaydi.



kontekst hajmi

Kontekst ko'paysa, model kuchliroq bo'lishi mumkin, lekin uni ishonchli o'qitish uchun ko'proq ma'lumot kerak.

Seminal manba: Rabiner (1989)

Rabiner, L.R. (1989). **A Tutorial on Hidden Markov Models and Selected Applications in Speech Recognition.** *Proceedings of the IEEE*, 77(2), 257–286.

Ahamiyati

- HMM uchun klassik va keng qo'llanadigan qo'llanma.
- Uch masala: baholash (Forward), dekodlash (Viterbi), o'rganish (Baum–Welch).
- Nutq, POS, bioinformatika — hammasi shu doirada.

Forward	**Viterbi**	Baum–Welch
baholash	dekodlash (bugun)	o'rganish

Muhokama savoli (5-amaliyotgacha o'ylang)

HMM teglash uchun **teglangan korpus** kerak. O'zbek tili uchun bunday korpus kam. Bu cheklovni qanday yengish mumkin? (Maslahat: 4-hafta, BERT.)

Bugungi maqsadlar bajarildi

1. ✓ **Keltirib chiqardik:** bigram MLE — $P(\text{kitob} \mid \text{men}) = 2/3$ (zanjir qoidasi + Markov taxmini).
2. ✓ **Hisobladik:** add-1 silliqlash ($P_{+1}(\text{o'qidim} \mid \text{non}) = 1/7$) va perplexity.
3. ✓ **Qo'lladik:** Markov zanjiri — $P(\mathrm{NN}, \mathrm{VB}) = 0.42$.
4. ✓ **Keltirib chiqardik:** Viterbi DP — $\delta(\mathrm{VB}, t{=}2) = 0.3402$, ketma-ketlik [NN, VB].

Ertaga 5-amaliyot: avtomatik to'ldirish va POS teglash quriladi

N-gram model + silliqlash → **m05** Autocomplete

HMM + Viterbi → **m05b** POSTagger

5-amaliyotda qurasiz

- `m05 Autocomplete`: bigram + add-1, perplexity;
- `m05b POSTagger`: HMM + `viterbi`;
- eng ehtimoliy davom va teg ketma-ketligi.

Birinchi assert (ushbu ma'ruzaning Viterbi slaydi bilan solishtiriladi):

```
assert abs(delta_VB_t2
           - 0.3402) < 1e-4
# yo'l == [NN, VB]
```

6-ma'ruzaga ko'prik: N-gram so'z ma'nosini bilmaydi

N-gram til modeli so'zlarni **alohida (diskret) token** sifatida sanaydi — shuning uchun ularning *ma'nosini* bilmaydi:

`kitob` va `kitobni`, yoki `film` va `kino` — model uchun butunlay **boshqa** tokenlar. Birida o'rgangan ehtimol boshqasiga o'tmaydi (ma'lumot bo'linib ketadi).

Word2Vec / CBOW: so'zlarni **zich vektorga** akslantiradi — ma'nosi yaqin so'zlar (`film`≈`kino`) fazoda yaqin, shuning uchun ma'lumot ular orasida **ulashiladi**.

Ya'ni: diskret tokenlardan → ma'noni ushlaydigan zich vektorlarga.

Adabiyotlar

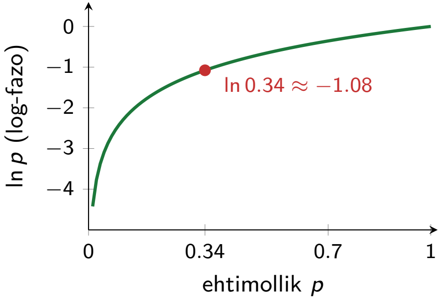
1. Rabiner, L.R. (1989). *A Tutorial on Hidden Markov Models and Selected Applications in Speech Recognition*. Proc. IEEE, 77(2), 257–286.
2. Jurafsky, D., & Martin, J. H. (2023). *Speech and Language Processing* (3-nashr). 3-bob: N-gram til modellari; 8-bob: POS va HMM.
3. Shannon, C. E. (1948). *A Mathematical Theory of Communication*. Bell System Technical Journal. (Til modellashtirish va entropiya ildizi.)
4. Chen, S. F., & Goodman, J. (1999). *An Empirical Study of Smoothing Techniques for Language Modeling*. Computer Speech & Language.
5. `nltk.lm` hujjatlari — N-gram til modellari API.

Logarifmik shakl

$$\log \delta_t(j) = \max_i \Big[\log \delta_{t-1}(i) + \log A(j \mid i) \Big] + \log B(o_t \mid j)$$

- Ko'paytirish $\longrightarrow$ **qo'shish**: kichik ehtimollar ko'paytmasi o'rniga logarifmlar yig'indisi $\Rightarrow$ underflow yo'q.
- log **monoton** o'suvchi $\Rightarrow$ arg max o'zgarmaydi: orqaga kuzatish (*backtrace*) bir xil yo'lni beradi.

Misol: $\delta_2(\mathrm{VB}) = 0.3402$ uchun $\log \delta_2(\mathrm{VB}) = \ln 0.3402 \approx -1.078$.



Kichik $p \Rightarrow$ katta manfiy log; tartib saqlanadi.

Ilova S2: Perplexity va entropiya bog'liqligi

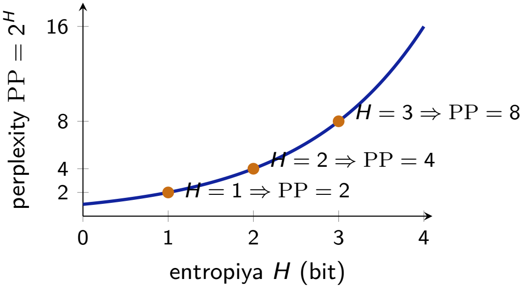
Asosiy bog'lanish

$$\mathrm{PP}(W) = 2^{H(W)}$$

$$H(W) = -\frac{1}{N}\sum_{i=1}^{N}\log_2 P(w_i \mid w_{i-1})$$

$H(W)$ — har bir token uchun o'rtacha noaniqlik (*entropy* — entropiya), bitlarda o'lchanadi.

- H kichik bo'lsa, model kamroq noaniqlikka ega.
- H oshsa, PP ham tez oshadi.
- 1 bit noaniqlik $\Rightarrow$ 2 ta teng ehtimolli tanlov.



Entropiya chiziqli oshsa, perplexity eksponensial oshadi. Shuning uchun past entropiya va past PP yaxshi model belgisi.